Spend Data
Integration Design Document

Date: March 2026
Status: Design confirmed (some prerequisites pending verification)

Executive Summary

This document defines the design policy for integrating Spend Data scattered across multiple systems and making it trustworthy. The current confidence level is approximately 50% (intuition-based), and this document identifies the root causes and the approach to resolving them.

Current Problems

Three compounding issues are creating the 50% confidence problem

Issue 1: Data preparation is ad-hoc and manual with no reproducibility

Issue 2: No formal rules defining which system to use for which vendor — decisions are undefined

Issue 3: No mechanism to verify data accuracy — correctness cannot be proven

Key Design Decisions

Three business decisions have been confirmed to resolve these issues.

Decision	Content	Rationale
Spend = Payment basis	PO amounts (contract totals) are excluded. Only records where payment has been executed are counted.	PO amounts represent pre-payment commitments and diverge from actual spend
SAP Concur takes precedence on conflict	When the same vendor has amounts in both Purchasing System and Concur, SAP Concur is adopted as the authoritative source	This project defines Spend on a payment basis, and Concur records actual payments
GL as external validation benchmark	GL cannot be used as vendor-level detail, but reconciling the integrated Spend total against GL account totals allows verification of omissions and double-counting	With three exclusive systems, there is no way to validate the whole without an external reference

Immediate Actions Required

The following four items must be verified against real data before implementation can proceed (Q1 and Q2 require design changes if they fail)

Q1: Confirm no erroneous Concur records exist for vendors already paid via the Purchasing System

Q2: Confirm no PO-amount records are mixed into the Purchasing System payment table

Q3: Confirm no duplicate overseas vendor records exist across the AP System and Purchasing System

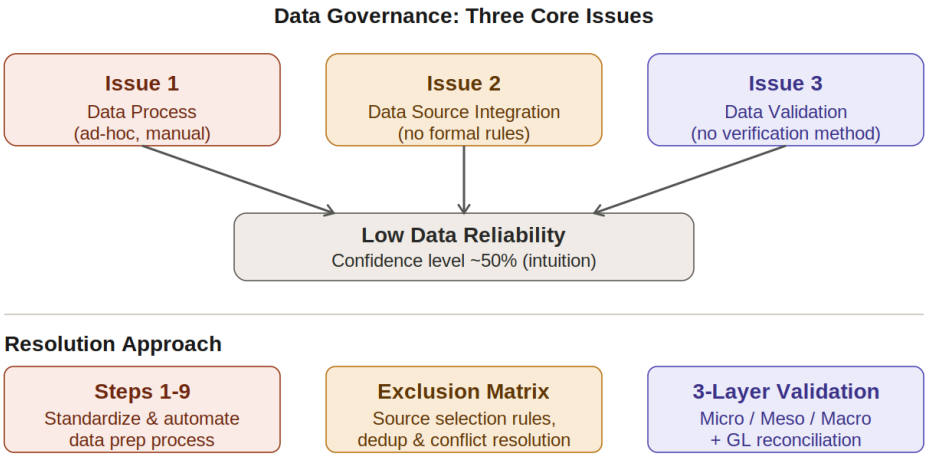
Q4: Confirm GL Data alignment on period, currency, account codes, and accrual vs. cash basis

Once confirmed, the short-term goal (~1 month) is to complete vendor ID matching and codify cleansing rules. The mid-term goal (~3 months) is to automate the pipeline. Monthly GL reconciliation as a quantitative confidence check will be established as ongoing steady-state operations.

1. Current Situation and Issues

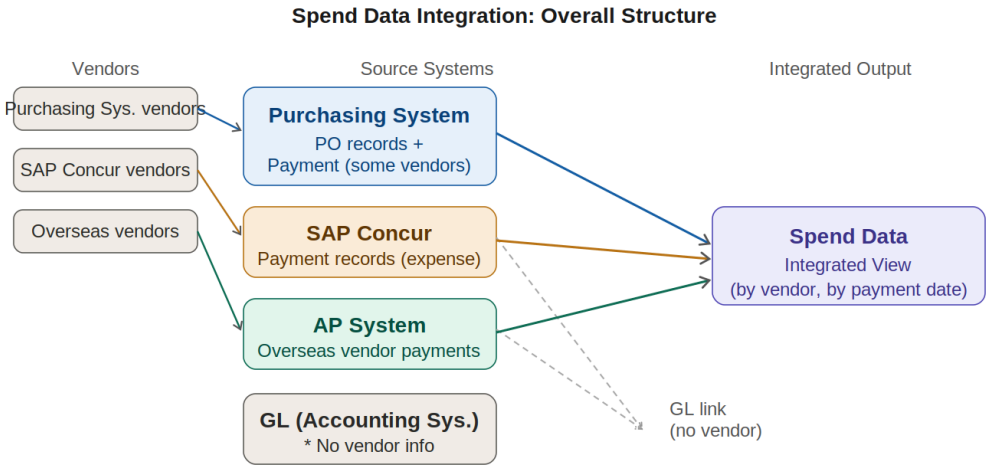
1.1 Why Is Spend Data Unreliable?

The current Spend Data is sourced from three different systems, but no rules exist defining how to integrate them. The preparation process depends on individual judgment, and there is no validation mechanism. These three factors combine to leave confidence at an intuition-based 50%.



1.2 The Reality of the Three Data Sources

Spend Data comes from the following three systems. Critically, each system covers different vendors and transaction types — they are principally exclusive to one another. This is not a problem of "the same data duplicated in three places" but rather "no rules exist defining who takes what from where."



System	Transactions Covered	Data Used
Purchasing System	Domestic vendor orders and some payments	Payment table only (PO table excluded)
SAP Concur	Payments to vendors not paid via Purchasing System	Payment records
AP System	Overseas vendor payments	Payment records
GL Data (Accounting)	Company-wide journal entries (no vendor info)	Validation use only — cannot use as detail

1.3 Dependencies Between the Three Issues

The three issues are not independent. The integration rules (Issue 2) must be defined before cleansing (Issue 1) can be designed, and Issues 1 and 2 must be complete before validation (Issue 3) can function. They must be resolved in the following order.

Resolution order: Issue 2 (confirm integration rules) -> Issue 1 (standardize data prep) -> Issue 3 (operationalize validation)

Ignoring this dependency and "fixing the prep process first" means rework when integration rules change.

2. Key Design Decisions

This chapter records the principal decisions confirmed for the Spend Data integration design. These are business agreements and form the prerequisites for technical implementation.

2.1 Spend Definition: Payment Basis

The Purchasing System contains two table types: a PO table (contract totals) and a payment table (actual payment records). This project uses the payment table only.

Example: 2-year contract with annual payments
PO table: 1 record (2-year total amount) -> Not used
Payment table: 2 records (Year 1 and Year 2 payment amounts) -> Used

2.2 Conflict Resolution: SAP Concur Takes Precedence

For vendors where a PO exists in the Purchasing System and a payment record exists in SAP Concur, the SAP Concur payment record is adopted as authoritative.

Source Selection Rules: Exclusion Matrix

Vendor Category	Purchasing System	SAP Concur	AP System
Purchasing System handles pay	Payment table (use)	No record	No record
PO in Purchasing / Payment in Concur	PO only (exclude)	Payment rec. (use)	No record
Overseas vendor	No record	No record	AP payment (use)

Conflict Resolution Rule:
When PO exists in Purchasing AND payment exists in Concur -> SAP Concur payment record takes precedence

2.3 Vendor ID Matching: Three-Phase Process

The same vendor may hold different IDs across systems. A matching table must be created prior to integration to map all records to a unified vendor ID. Matching cannot be completed by technology alone — the question "are these two vendor IDs the same entity?" always requires human judgment from procurement and finance teams.

Vendor ID Matching: 3-Phase Flow

Step 3 Discovery Phase -- Candidate Identification

Extract vendor ID + name lists from all systems / Identify candidates: exact match, partial match, name variants

Output: Candidate table (System A ID / System B ID / Match type)

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Business Confirmation
Human judgment required -- only manual step

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Step 4 Definition Phase -- Build Mapping Master Table

Assign unified vendor ID based on confirmed results / Create and maintain ID mapping table per system

Purchasing Sys. ID	SAP Concur ID	AP System ID	Unified ID / Name
VEN-001	CON-0045	--	M-001 / ABC Corp
--	--	AP-0078	M-002 / XYZ Ltd

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Step 5 Application Phase -- Apply to Source Data

JOIN mapping table to replace system IDs with unified vendor ID / Flag unmapped vendors as UNMAPPED and quaran

Ongoing process: Run Step3 -> Business confirmation -> Step4 update cycle for every new vendor added

2.4 GL Data: External Validation Benchmark

GL Data cannot be used as vendor-level detail because it lacks vendor information. However, by reconciling "integrated Spend total = GL account total," it serves as the only external benchmark capable of simultaneously verifying omissions and double-counting.

The following must be confirmed via Q4 before GL can be used as a benchmark (see Next Actions)

Period close dates are consistent between the three systems and GL

FX conversion rates and conversion timing are consistent

Target GL accounts fully correspond to Spend categories with no gaps or overlaps

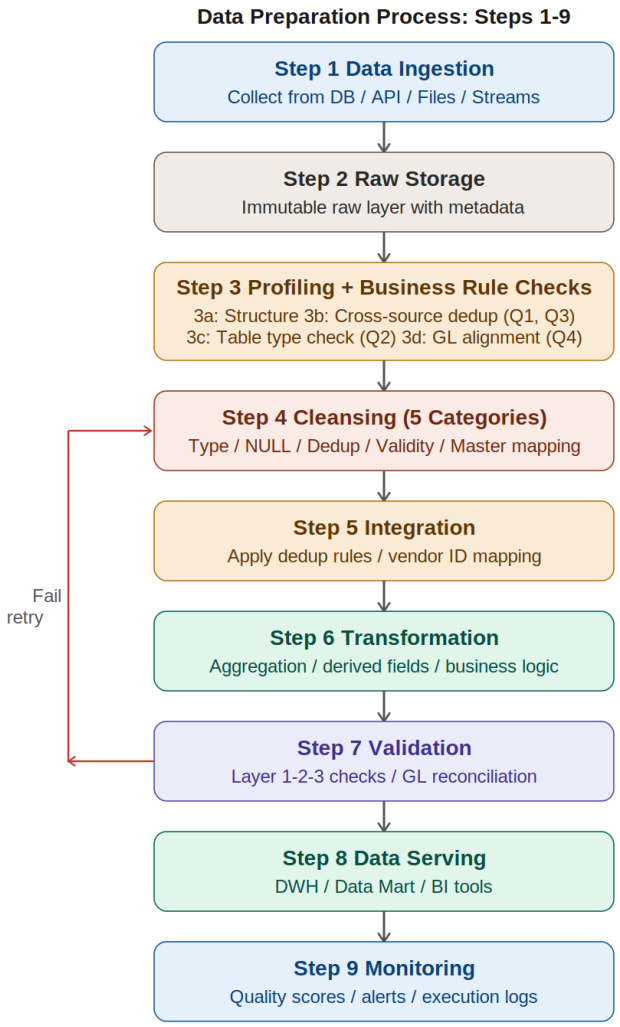
How to handle accrual basis (GL) vs. cash basis (Spend) differences

3. Implementation Approach

The concrete implementation approach for resolving Issues 1, 2, and 3 is described below in dependency order.

3.1 Data Preparation Process Standardization (Issue 1)

Replace the current ad-hoc process with the standard Steps 1-9 flow. Steps 3 and 4 receive project-specific extended definitions.



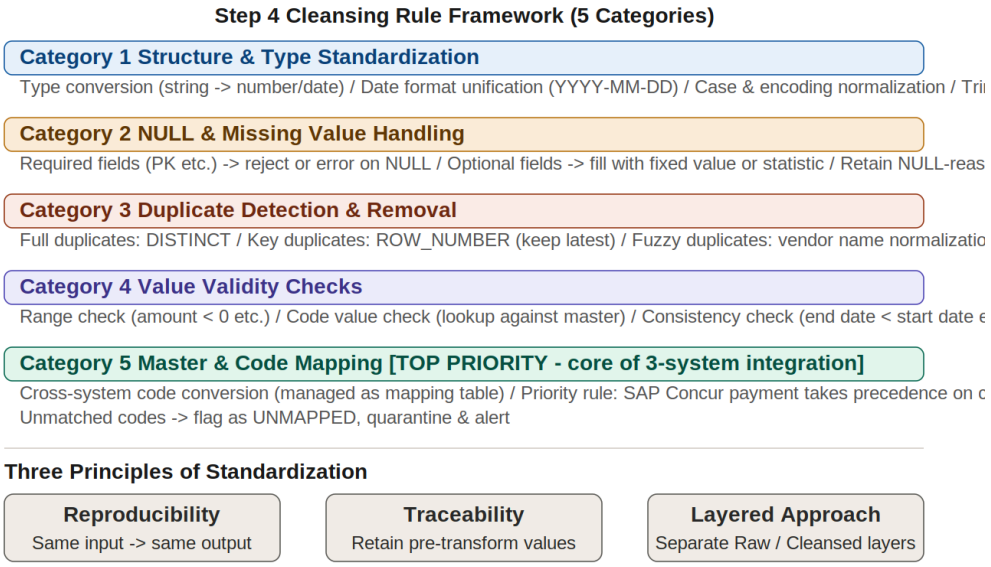
Step 3 Extension: Add Business Rule Verification

In addition to standard profiling (NULL rates, types, distributions), the following four sub-steps specific to Spend Data integration are required. The execution order must be: vendor matching creation -> Q1 -> Q2 -> Q3 -> Q4.

Sub-step	Activity	Related Q
3a Structure profiling	NULL rate, types, row counts, distributions (standard profiling)	--
3b Cross-source dedup scan	Confirm the same vendor does not appear in multiple systems	Q1, Q3
3c Table type check	Confirm no PO-amount records are mixed into payment data	Q2
3d GL alignment check	Confirm GL alignment on period, currency, and account codes	Q4

Step 4 Cleansing Rules: Standardized Across 5 Categories

Explicitly define all cleansing rules as code to eliminate ad-hoc manual judgment. Category 5 (code mapping) is the core of this integration and the highest priority.



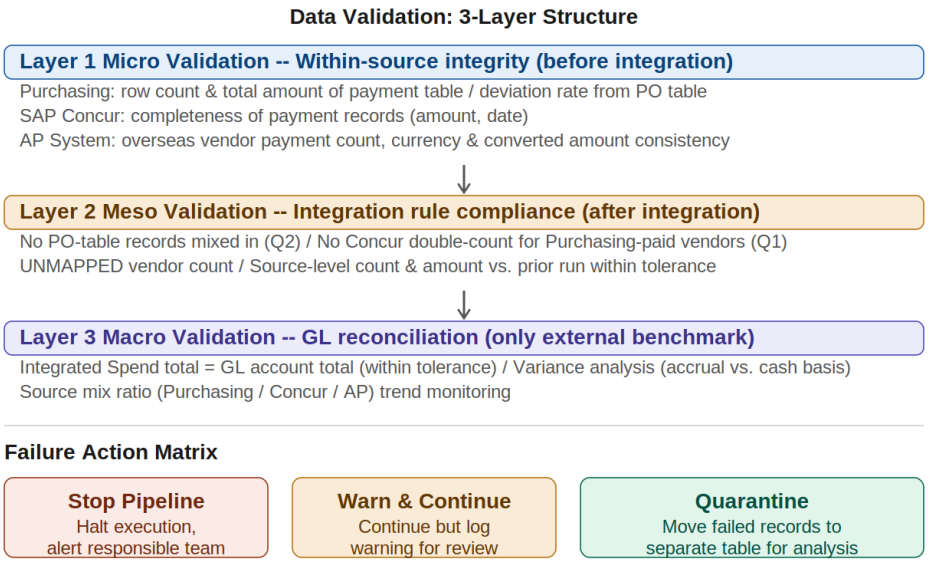
3.2 Integration Processing Flow (Issue 2)

The three sources are integrated following the per-vendor source selection rules in the order below.

1. Extract from Purchasing System payment table only (PO table excluded)
2. Extract SAP Concur payment records (exclude vendors already paid via Purchasing System)
3. Extract AP System overseas vendor payments
4. Deduplication check and removal: reconcile by vendor x payment date x amount
5. Reconcile integrated total against GL Data
6. Output as Spend Data integrated view by vendor, payment date, and source

3.3 Three-Layer Validation Structure (Issue 3)

Because the three systems cover exclusively different vendors and transaction types, direct cross-system reconciliation cannot detect omissions or double-counting. Validation is therefore structured in three layers.



Layer	Timing	Key Checks	What It Ensures
Layer 1 Micro validation	Before integration	Per-system row count, total amount, and completeness checks	Source data quality
Layer 2 Meso validation	After integration	No PO-amount contamination, no double-counting, UNMAPPED count	Integration rule accuracy
Layer 3 Macro validation	Monthly steady-state	Integrated Spend total = GL total (within tolerance)	Absence of omissions and double-counting

3.4 Automation Policy

Because the current process is largely manual, insufficient time remains for validation. The following approach will reduce "execution" work and allow focus on "judgment."

- Codify all cleansing rules so that results are reproducible regardless of who runs the process
- Run Step 7 validation automatically; humans focus on investigating failures and deciding responses
- Automate vendor matching candidate extraction to minimize manual confirmation effort

4. Next Actions and Confirmation Items

This chapter is an action list defining who does what by when. Q1 and Q2 require design changes if they fail — both must be completed before implementation begins.

4.1 Real Data Verification (Q1-Q4)

Q1 and Q2 are blockers: failures require revision of the integration rules in Chapter 2

Item	Check Content	Expected Result	If Fails	Owner	Status
Q1	Any erroneous Concur records for vendors already paid via Purchasing System?Join key: Vendor ID x payment date x amount	Matching records = 0	Quarantine matching records; confirm authoritative system with business side; define exclusion rule in Step 4	--	Not started
Q2	Any PO-amount records mixed into Purchasing System payment table?Also check for NULL or future-dated payment dates	Payment flag only; all payment dates are real past dates	Identify contaminated records; define exclusion condition in Step 4 cleansing rules (Category 4)	--	Not started
Q3	Any duplicate overseas vendor records across AP System and Purchasing System?Also check for name-matching duplicates across different IDs	Duplicate vendors = 0; name-match candidates = 0	Define authoritative source rule for duplicate vendors; update matching table; rerun check	--	Not started
Q4	Are GL alignment conditions consistent?Period, currency, account codes, accrual vs. cash basis	All conditions aligned; tolerance range defined	Define correction rules per misaligned item, or limit GL reconciliation scope accordingly	--	Not started

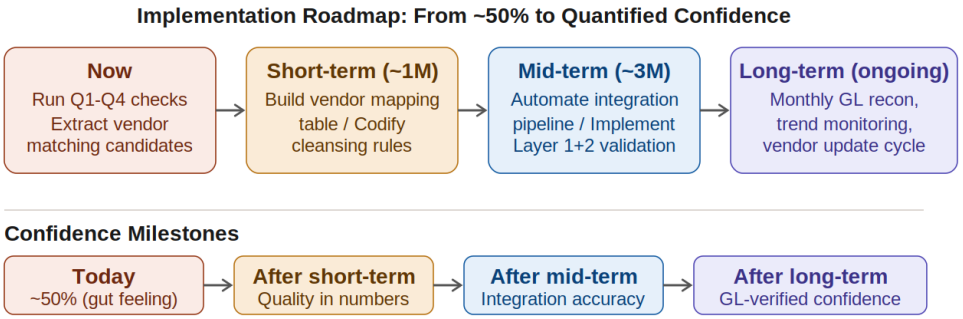
4.2 Business Confirmations for Vendor Matching Table

After Step 3b (matching candidate extraction) is complete, the following must be confirmed with procurement and finance teams. This is the only step in the process that cannot be replaced by technology — human judgment is required.

- Approval of the candidate list of presumed same-entity vendors
- ID assignment convention for unified vendor IDs where system IDs differ
- Agreement on canonical vendor name when name variants are identified

4.3 Phase-by-Phase Implementation Roadmap

Progress from the current intuition-based "~50% confidence" to a state where reliability is expressed as a concrete number — in stages.



Phase	Timeline	Key Activities	Target State
Now	Immediate	Run Q1-Q4 check queries; extract vendor matching candidates	Prerequisites for integration confirmed
Short-term	~1 month	Build vendor matching table (incl. business confirmation); codify cleansing rules	Source quality measurable in numbers
Mid-term	~3 months	Automate integration pipeline; implement Layer 1 and 2 validation	Integration rule accuracy verifiable
Long-term (ongoing)	Steady-state	Monthly GL reconciliation; trend monitoring; vendor matching update cycle	GL reconciliation quantitatively confirms omissions and double-counting

4.4 Ongoing Operational Cycle

The following three activities form the continuous data governance cycle. Building the pipeline once is not the end — the cycle must keep running with every new vendor addition or system change.

Cycle	Frequency	Content	Owner
Vendor matching updates	On new vendor addition	Step 3 candidate extraction -> business confirmation -> matching table update	--
GL reconciliation	Monthly	Reconcile integrated Spend total against GL total; alert if variance exceeds tolerance	--
Delta monitoring	Each pipeline run	Log period-on-period row count and amount changes; auto-detect anomalies	--

Appendix: Technical Reference

This chapter is a reference for technical staff. Managers should consult as needed.

A. Q1-Q4 Check Query Requirements

The following are the requirements for the business rule verification checks carried out as part of Step 3 (profiling). Actual SQL is implemented based on these requirements.

Prerequisite: Q1 and Q3 require a vendor matching table to exist before execution. Without it, the same vendor under different IDs will be missed.

Common note 1: All queries must include the target period as a filter condition. Without it, historical clean data contaminates the results.

Common note 2: Q1-Q4 results (row counts and amounts) must be saved as Step 3 output logs. They will be reused as GL reconciliation baseline values in Step 7.

Q1 Erroneous Registration in SAP Concur

Confirm that vendors whose payments are fully handled by the Purchasing System are not also registered in SAP Concur. If such cases exist, the same payment will be double-counted from both systems during integration.

Item	Detail
Purpose	Confirm that vendors paid via the Purchasing System do not also have payment records in SAP Concur
Join key	Vendor ID (unified ID via matching table) x payment date x payment amount
Verification steps	1. Retrieve the list of vendor IDs present in the Purchasing System payment table2. Check whether those vendor IDs appear in SAP Concur payment records3. If found, check whether records with matching payment date and amount exist
Expected result	Matching records = 0
If fails	Quarantine matching records to a Quarantine table. Confirm which system is authoritative with the business side and define an exclusion rule in Step 4

Q2 PO Amount Contamination in Purchasing Payment Table

Confirm that the Purchasing System payment table does not contain PO-amount records (contract totals) mixed in alongside actual payment records. Contamination causes the payment-basis aggregation to be significantly overstated.

Item	Detail
Purpose	Confirm that no PO-amount records are mixed into the payment table
Identification key	Record-type flag (PO / Payment) or equivalent column value
Verification steps	1. Check the distribution of record-type flag values in the payment table2. Check for records with a NULL or future-dated payment date3. Check whether the sum of

Item	Detail
	payment records for a given PO ID matches the PO amount in the PO table (a match suggests the PO amount itself may have been inserted as a payment record)
Expected result	Record-type flag = "Payment" only; all payment dates are real past dates
If fails	Identify the contaminated records and define their exclusion condition explicitly in Step 4 cleansing rules (Category 4: Validity Checks)

Q3 Overseas Vendor Overlap: AP System x Purchasing System

Confirm that the same overseas vendor is not registered in both the AP System and the Purchasing System with payment records in each.

Item	Detail
Purpose	Confirm that payments to the same overseas vendor are not duplicated across both systems
Join key	Vendor ID (unified ID via matching table) x payment date x payment amountNote: also check cases where IDs differ but names are the same (matching table gaps)
Verification steps	1. Retrieve the list of overseas vendor IDs in the Purchasing System payment table2. Check whether those IDs appear in the AP System vendor master or payment records3. If found, check whether records with matching payment date and amount exist in both systems4. Run name-matching to detect vendors with similar names under different IDs (matching table gap detection)
Expected result	Duplicate vendors = 0; name-match candidates = 0
If fails	Define the authoritative source for duplicate vendors as a rule; add the exclusion filter to Step 4. If name-match candidates are found, update the matching table and rerun the check

Q4 GL Data Alignment Confirmation

Confirm that the aggregation conditions of the three source systems and GL are aligned before using GL Data as the Layer 3 macro validation benchmark in Step 7. Running reconciliation without this confirmation risks misclassifying condition-based variance as "data errors."

Check Item	Check Content	If Misaligned
1. Period	Confirm that monthly close dates are the same across the three systems and GL	Define a correction rule establishing which timing is the reference standard
2. Currency / FX	Confirm that FX conversion rates and conversion timing (transaction-date rate, month-end rate, etc.) are consistent	Define the conversion variance as part of the tolerance range, or unify the conversion method
3. Account codes	Identify the GL account codes corresponding to Spend categories and create a mapping table. Check for any gaps or overlaps in coverage	Limit GL reconciliation to the specific confirmed account codes
4. Accrual vs. cash	If GL records on an accrual basis (expense recognition date), confirm how the difference from cash-basis Spend (payment execution date) is handled	Define accrual-to-cash differences (accrued liabilities, etc.) as part of the tolerance range, or agree with finance on a cash-basis GL extraction condition

Q4 expected result: All four items above are aligned AND a GL reconciliation tolerance range (e.g., within +/-1%) is defined

Even if not all conditions align, GL can still be used as a validation benchmark as long as correction rules or scope limitations are agreed upon for each misaligned item

B. Step 7 Validation Checklist (All Items)

Validation checklist with implementation timing targets by priority.

Priority	Target	Check Item	Validation Logic
Required	This week	Total row count	Pre/post-processing row counts match
Required	This week	Abnormal row count drop	Alert if variance vs. prior run exceeds threshold
Required	This week	Per-source row count reconciliation	Record and compare each source's row count
Required	This week	NULL count ceiling	NULL rate in required fields below threshold
Required	This week	Total amount reconciliation	Aggregated amount matches source
Required	This week	Cross-source reconciliation	Record count of value mismatches across systems
Required	2-3 weeks	Foreign key reference check	No codes absent from master table
Recommended	1 month	Aggregation granularity	Line-item total = header total
Recommended	1 month	Outlier detection	Values beyond mean +/-3 sigma flagged for review
Recommended	1 month	Period-on-period amount variance	Total amount within tolerance vs. prior run
Maturity	Continuous	New / deleted key detection	Alert on keys added or removed vs. prior run
Maturity	Continuous	Time-series anomaly detection	Deviation from weekly/monthly trend patterns

C. Failure Action Matrix

Defining failure responses in advance prevents ad-hoc judgment and reintroduces the human-dependency problem the process redesign is meant to eliminate.

Failure Type	Action	Example Trigger	Decision Criteria
Stop pipeline	Halt execution; alert responsible team	Large total amount deviation; mass NULL in required keys	Variance exceeds 2x tolerance, or NULL rate > threshold
Warn and continue	Continue processing; log warning for review	Minor row count variance; small number of outliers	Within tolerance but statistically significant vs. prior run

Failure Type	Action	Example Trigger	Decision Criteria
Quarantine	Move failed records to separate table for later analysis	UNMAPPED vendors; unrecognized code values	Record-level quality issue that can be specifically identified

This document is the agreed design specification for the design phase and will be updated to reflect actual implementation outcomes upon completion.